

Lucas Zheng

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SKILLS

- Programming languages: Java, Python, C++
- Systems: Microcontrollers, Linux, shell, Oculus VR
- Robotics: ROS, Gazebo, control theory
- Fullstack: Flutter, Dart, Kotlin, Unity, HTML, CSS, JavaScript, TypeScript, React, Electron, Flask, Node, Next, SQL, NGINX, Docker
- Problem solving: math, programming, and robotics competitions
- Additional expertise: data structures and algorithms, Scheme, LaTeX, Git, AWS

PROJECTS

FlySafe April 2023 - January 2024

- Invented an affordable, modern AGL plane flight altimeter using ESP32 with LIDAR and WebSocket technology. The device helps amateur pilots improve their landing skills by delivering real-time visual and audio feedback integrated with aircraft sound systems. Designed for easy and secure mounting on the plane wing.
- Developed an intuitive web app for real-time data monitoring, device configuration, and post-flight data analysis, providing pilots with insights to improve performance.

EZ-Check November 2022 - June 2024

- Developed a novel IoT authentication system to secure VCHS maker lab tools, leveraging off-the-shelf ESP32 for seamless integration with electronics and backend systems, including designing 3D enclosures and performing board soldering for a complete solution. Designed to provide enhanced security and controlled access to school's expensive machines.
- Built a full-stack platform for administrators to manage students, supervisors, and machines, streamlining the operation of the maker lab.

EZ-Find November 2022 - June 2024

- Designed an innovative inventory management app used across school storages for efficient item filling and locating.
- Developed full-stack solution that simplifies the management of users, organizations, storages, and inventory by integrating a QR-code system and automatic label printing functionality.

Google Tensorflow Lite for Microcontrollers Challenge May 2021 - June 2021

- Developed an interactive AI-powered Arduino pet by integrating Arduino Nano, BLE, IMU, and Tensorflow, enabling real-time emotion tracking. The project also involved creating a website to monitor the pet's emotions live.
- Self-taught in TinyML through EdX Harvard courses.

EXPERIENCE

FRC Robotics Team 3256: Software Developer / Software Co-lead October 2020 - June 2023 / June 2023 - May 2024

- Qualified for the World Championship in 2022, 2023, and 2024.
- Developed robotics solutions including dynamic path generation with Bezier curves and Dijkstra's algorithm, LED animation library, match scouting app, and programming/simulation of robotic subsystems.
- Led training sessions by delivering lectures on complex topics, reviewing code commits, and mentoring peers to elevate team performance.

Avatar XPRIZE Team Avatar Quest: Software Developer August 2021 - September 2021

- Qualified for the semifinals (top 38), being the only pre-collegiate team to advance. We developed a humanoid with telepresence capabilities.
- Contributed to the development of software for wrist control and gripper functions.

Wildfire XPRIZE Team Wildfire Quest: Team Lead / Advisor July 2022 - July 2023 / August 2023 - Present

- Qualified for Track B (top 30). We designed a system for the rapid detection and suppression of small wildfires over a large area. Appointed to lead a team of 9 at the inception of the project, overseeing all aspects of project development and helped deliver the technical submission.

VCHS Research and Development: Honors Fellow

Underwater360 May 2023 - August 2024

- Developed software for a controllable underwater 360° camera system to enhance marine biology education through immersive experiences.

- Modified an Insta360 camera by integrating it with a Khadas VIM4 SBC and creating an Android-based solution (Kotlin) to interface with the camera.
- Developed a full-stack website for remote camera operation, featuring interactive 360° live streaming and customizable capture settings.

MBARI AUV Software Developer

June 2023 - August 2023

- Contributed to the development of an autonomous underwater vehicle capable of delivering experiments up to 1 km below the surface.
- Implemented underwater localization using ROS2 in the Gazebo simulator, integrating Kalman Filter sensor fusion with USBL beacon, IMU, and dynamics data (ongoing effort).

EZ-Gimbal

August 2022 - October 2022

- Co-developed a controllable 3-axis auto-stabilizing gimbal project: modified a RoninSC Gimbal with a serial inverter and SBUS, developed a ROS node for robot integration, and streamed stereoscopic camera to a VR headset.

Quest Institute RoboQuest

Junior High Program Developer and TA

October 2019 - May 2021

- Assisted in the development and testing of program curriculum and robots. Taught students to build and program robots using micro:bit and block coding through virtual classes during COVID19.

Junior High Program Intern and TA Trainer

June 2022 - August 2022

- Built a robot using Jetson Nano and ROS2 to explore potential for a new school program and developed a website for remote teleoperation and live camera streaming.
- Taught robotics concepts and the RoboQuest curriculum, and mentored incoming teaching assistants.

Experimental Donkey Car Team Research and Development Lead

May 2021 - September 2021

- Trained a virtual car to race using Behavioral Cloning AI to compete in the Virtual Donkey Car Competition.
- Developed software for a Raspberry Pi-based tank to autonomously navigate a track with Tensorflow, participating in the real-life Donkey Car Competition.

High School Program Alpha-tester and helper

November 2020 - October 2021

- Enhanced kit capabilities by developing software and a control framework to integrate 2D LIDAR mapping, immersive VR teleoperation, and Tensorflow.

AWARDS

USA Computing Olympiad: 2023 US Open Gold Division #154 out of 880. Perfect score on Bronze and Silver contests.

USA Math Olympiad: 4x American Invitational Mathematics Exam (AIME) qualifier, 2023 school highest AIME scorer (score of 7), 2x AMC distinction (top 5%)

Math League: 2023 International Math League 5th place, 2x North California Regional 2nd place

EDUCATION

University of California, Berkeley

May 2024 - May 2028

Bachelor's, Computer Science

GPA: 4.0

Sound and Music Computing with CNMAT Technologies (A+), Structure and Interpretation of Computer Programs (A+), Data Structures (in progress), Discrete Mathematics and Probability Theory (in progress), Abstract Linear Algebra (in progress)

Valley Christian High School (VCHS)

August 2020 - May 2024

Unweighted GPA: 4.0, weighted GPA: 4.71 | SAT: 1550 | Math Team Co-President, Arduino Club Co-President

Linear Algebra Honors, Multivariable Calculus Honors, Advanced Data Analysis Honors, AP CS A

DeAnza College

April 2021 - August 2022

Intro Linux/Unix (A), Intro C++ (A+), Intermediate C++ (A)

REFERENCES

Available upon request